

Multiple linear regression

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Grading the professor

Many college courses conclude by giving students the opportunity to evaluate the course and the instructor anonymously. However, the use of these student evaluations as an indicator of course quality and teaching effectiveness is often criticized because these measures may reflect the influence of non-teaching related characteristics, such as the physical appearance of the instructor. The article titled, “Beauty in the classroom: instructors’ pulchritude and putative pedagogical productivity” by Hamermesh and Parker found that instructors who are viewed to be better looking receive higher instructional ratings.

Here, you will analyze the data from this study in order to learn what goes into a positive professor evaluation.

Getting Started

Load packages

In this lab, you will explore and visualize the data using the **tidyverse** suite of packages. The data can be found in the companion package for OpenIntro resources, **openintro**.

Let’s load the packages.

```
library(tidyverse)
library(openintro)
library(GGally)
```

This is the first time we’re using the **GGally** package. You will be using the **ggpairs** function from this package later in the lab.

The data

The data were gathered from end of semester student evaluations for a large sample of professors from the University of Texas at Austin. In addition, six students rated the professors’ physical appearance. The result is a data frame where each row contains a different course and columns represent variables about the courses and professors. It’s called **evals**.

```
glimpse(evals)

## Rows: 463
## Columns: 23
## $ course_id    <int> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 1~
## $ prof_id      <int> 1, 1, 1, 1, 2, 2, 2, 3, 3, 4, 4, 4, 4, 4, 4, 4, 5, 5, ~
## $ score        <dbl> 4.7, 4.1, 3.9, 4.8, 4.6, 4.3, 2.8, 4.1, 3.4, 4.5, 3.8, 4~
## $ rank         <fct> tenure track, tenure track, tenure track, tenure track, ~
```

```
## $ ethnicity    <fct> minority, minority, minority, minority, not minority, no-
## $ gender       <fct> female, female, female, female, male, male, male, male, ~
## $ language     <fct> english, english, english, english, english, english, en-
## $ age          <int> 36, 36, 36, 36, 59, 59, 59, 51, 51, 40, 40, 40, 40, 40, ~
## $ cls_perc_eval <dbl> 55.81395, 68.80000, 60.80000, 62.60163, 85.00000, 87.500~
## $ cls_did_eval <int> 24, 86, 76, 77, 17, 35, 39, 55, 111, 40, 24, 24, 17, 14,~
## $ cls_students <int> 43, 125, 125, 123, 20, 40, 44, 55, 195, 46, 27, 25, 20, ~
## $ cls_level    <fct> upper, upper, upper, upper, upper, upper, upper, upper, ~
## $ cls_profs    <fct> single, single, single, single, multiple, multiple, mult~
## $ cls_credits  <fct> multi credit, multi credit, multi credit, multi credit, ~
## $ bty_f1lower  <int> 5, 5, 5, 5, 4, 4, 4, 5, 5, 2, 2, 2, 2, 2, 2, 2, 2, 7, 7,~
## $ bty_f1upper  <int> 7, 7, 7, 7, 4, 4, 4, 2, 2, 5, 5, 5, 5, 5, 5, 5, 5, 9, 9,~
## $ bty_f2upper  <int> 6, 6, 6, 6, 2, 2, 2, 5, 5, 4, 4, 4, 4, 4, 4, 4, 4, 9, 9,~
## $ bty_m1lower  <int> 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 7, 7,~
## $ bty_m1upper  <int> 4, 4, 4, 4, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 6, 6,~
## $ bty_m2upper  <int> 6, 6, 6, 6, 3, 3, 3, 3, 3, 2, 2, 2, 2, 2, 2, 2, 2, 6, 6,~
## $ bty_avg      <dbl> 5.000, 5.000, 5.000, 5.000, 3.000, 3.000, 3.000, 3.333, ~
## $ pic_outfit   <fct> not formal, not formal, not formal, not formal, not form~
## $ pic_color    <fct> color, color, color, color, color, color, color, color, ~
```

We have observations on 21 different variables, some categorical and some numerical. The meaning of each variable can be found by bringing up the help file:

```
?evals
```

Exploring the data

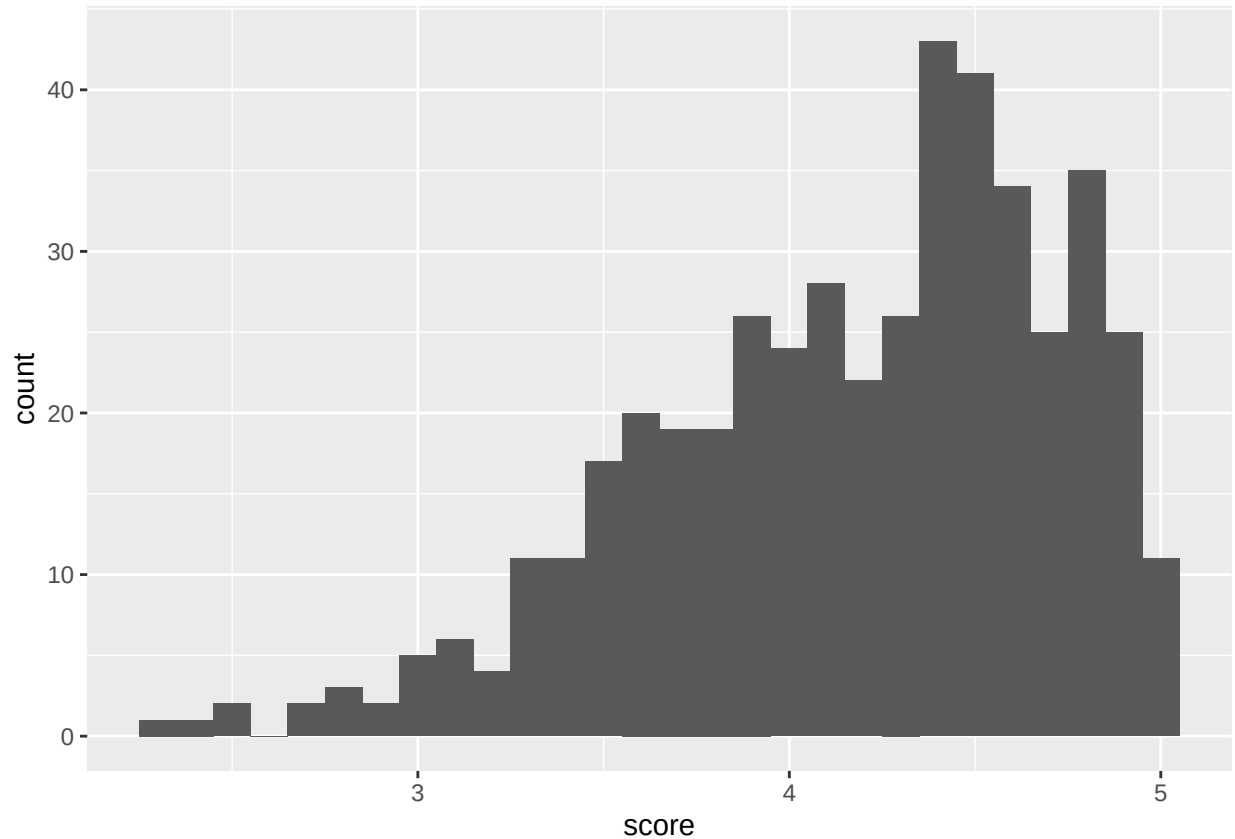
1. Is this an observational study or an experiment? The original research question posed in the paper is whether beauty leads directly to the differences in course evaluations. Given the study design, is it possible to answer this question as it is phrased? If not, rephrase the question.

Insert your answer here This is an observational study. There is no treatment done by the researchers which means its not possible to isolate different variables and determine which are confounding variables. A better rephrased question would be whether there is an association between a professors physical appearance and their course eval score.

2. Describe the distribution of `score`. Is the distribution skewed? What does that tell you about how students rate courses? Is this what you expected to see? Why, or why not?

Insert your answer here

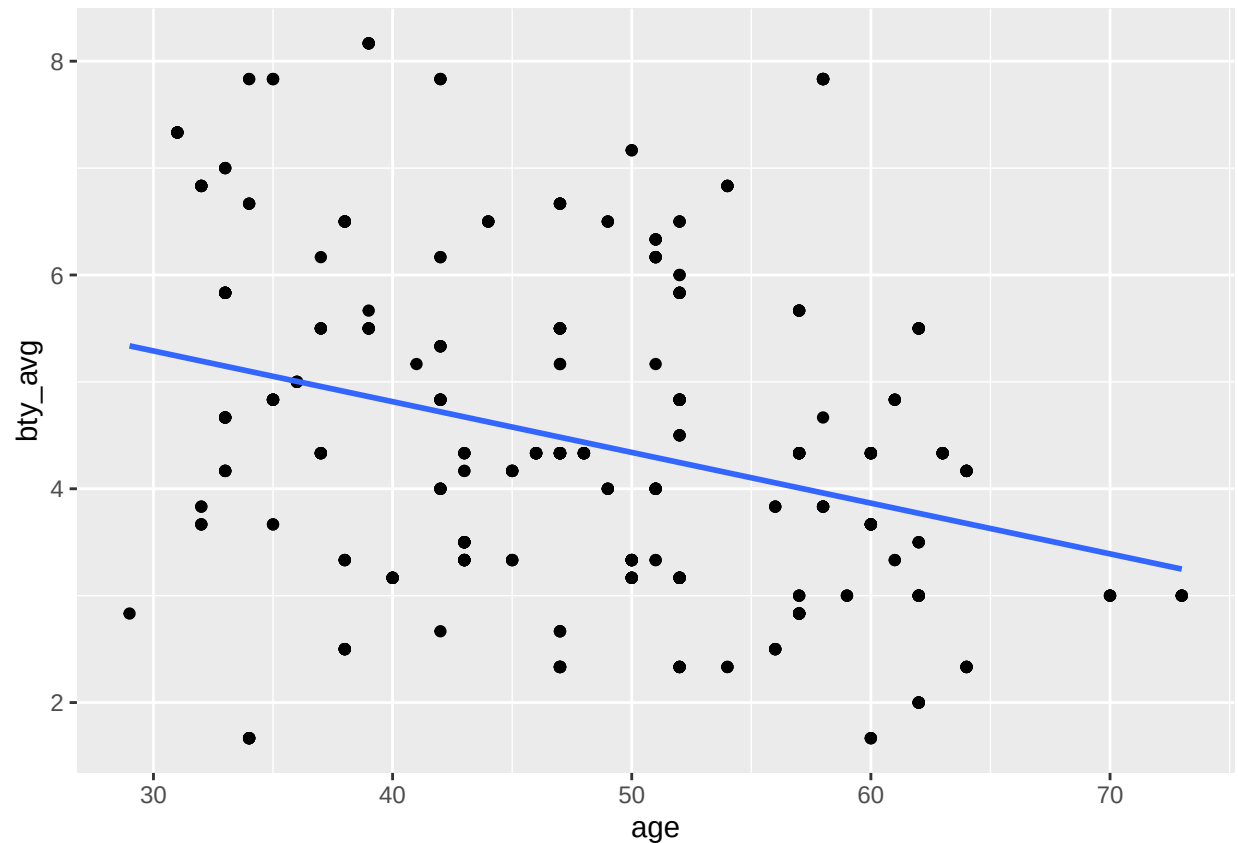
```
ggplot(evals, aes(x = score)) + geom_histogram(binwidth = 0.1)
```



The distrution is left skewed. It is whats expected because most dissatisfied students don't tend to leave negative ratings. 3. Excluding `score`, select two other variables and describe their relationship with each other using an appropriate visualization.

Insert your answer here

```
ggplot(evals, aes(x = age, y = bty_avg)) + geom_point() + geom_smooth(method = "lm", se = FALSE)
```

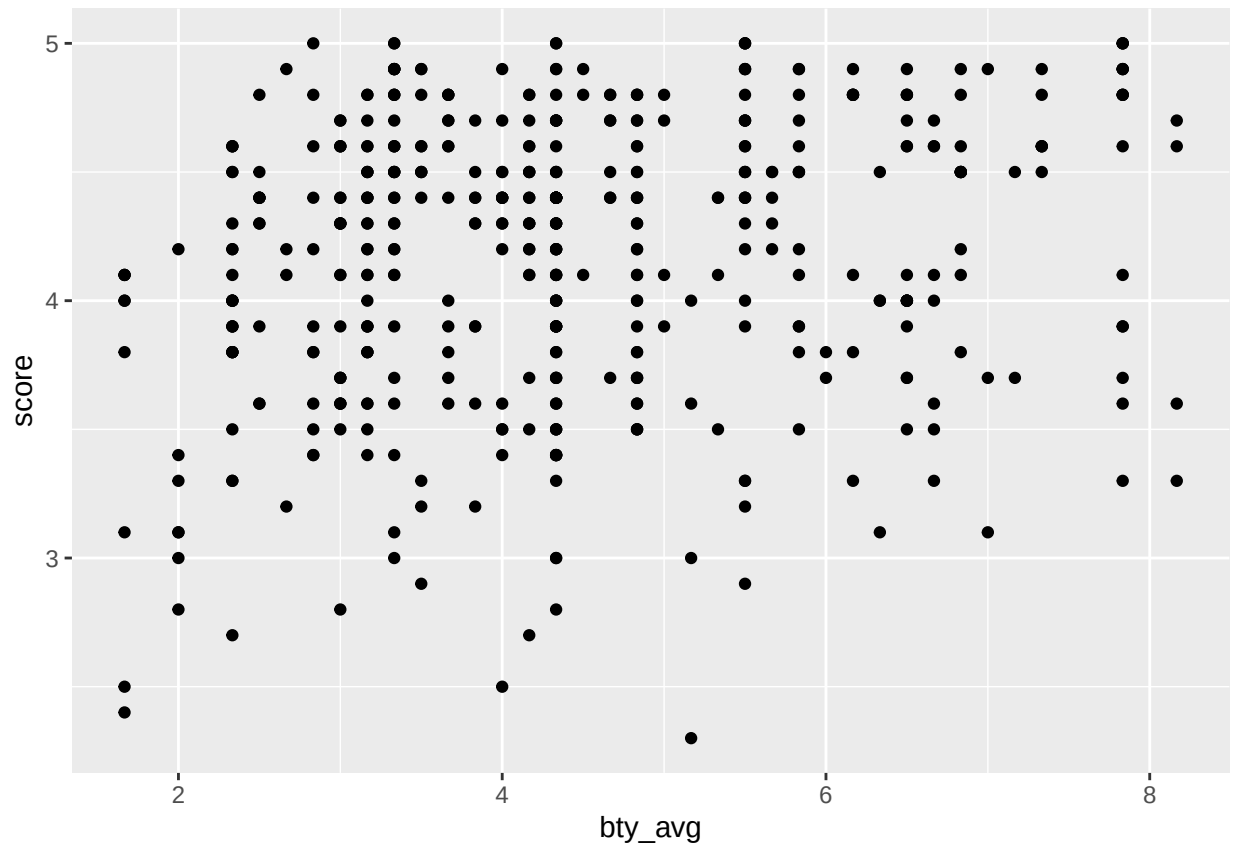


age and bty_avg shows a negative relationship. older professors tend to receive slightly lower beauty ratings on average. This is consistent with societal beauty norms but it is important to know that the relationship is not strong.

Simple linear regression

The fundamental phenomenon suggested by the study is that better looking teachers are evaluated more favorably. Let's create a scatterplot to see if this appears to be the case:

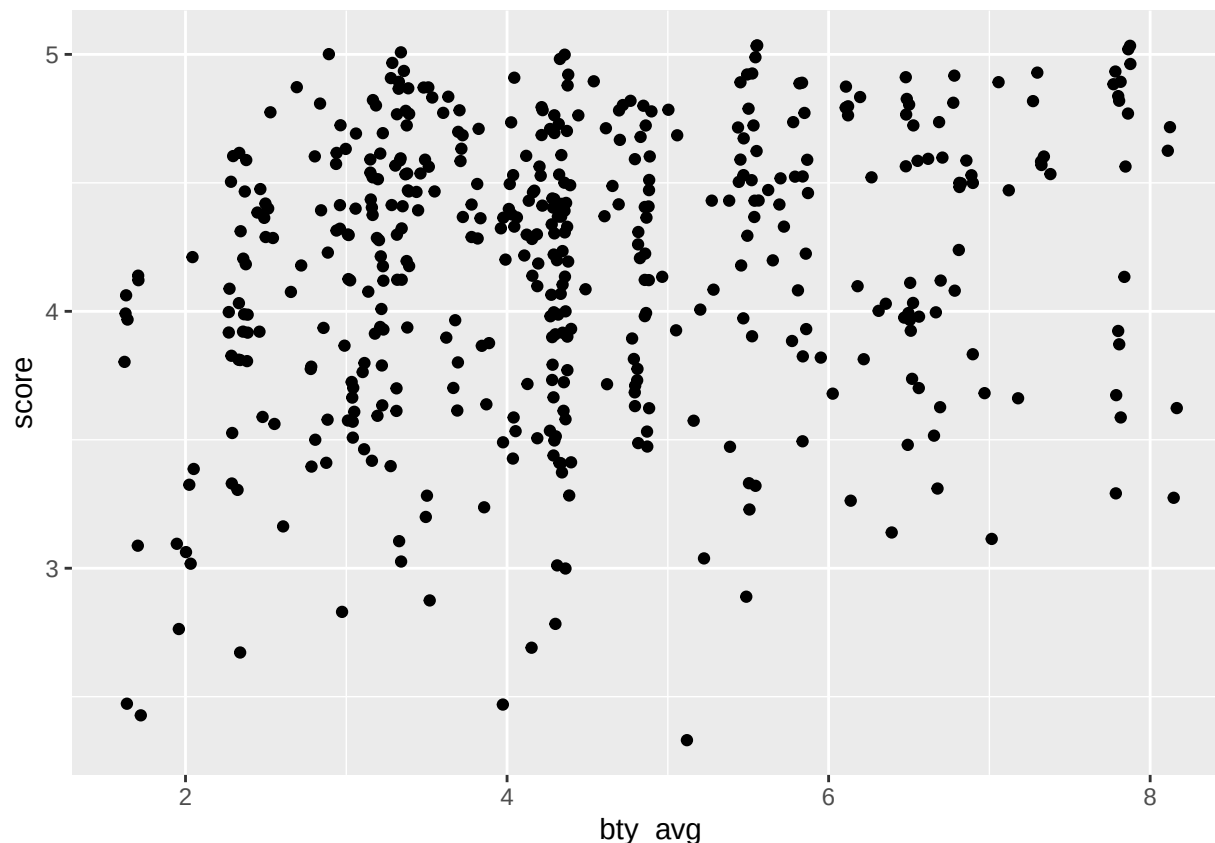
```
ggplot(data = evals, aes(x = bty_avg, y = score)) +  
  geom_point()
```



Before you draw conclusions about the trend, compare the number of observations in the data frame with the approximate number of points on the scatterplot. Is anything awry?

4. Replot the scatterplot, but this time use `geom_jitter` as your layer. What was misleading about the initial scatterplot?

```
ggplot(data = evals, aes(x = bty_avg, y = score)) +  
  geom_jitter()
```



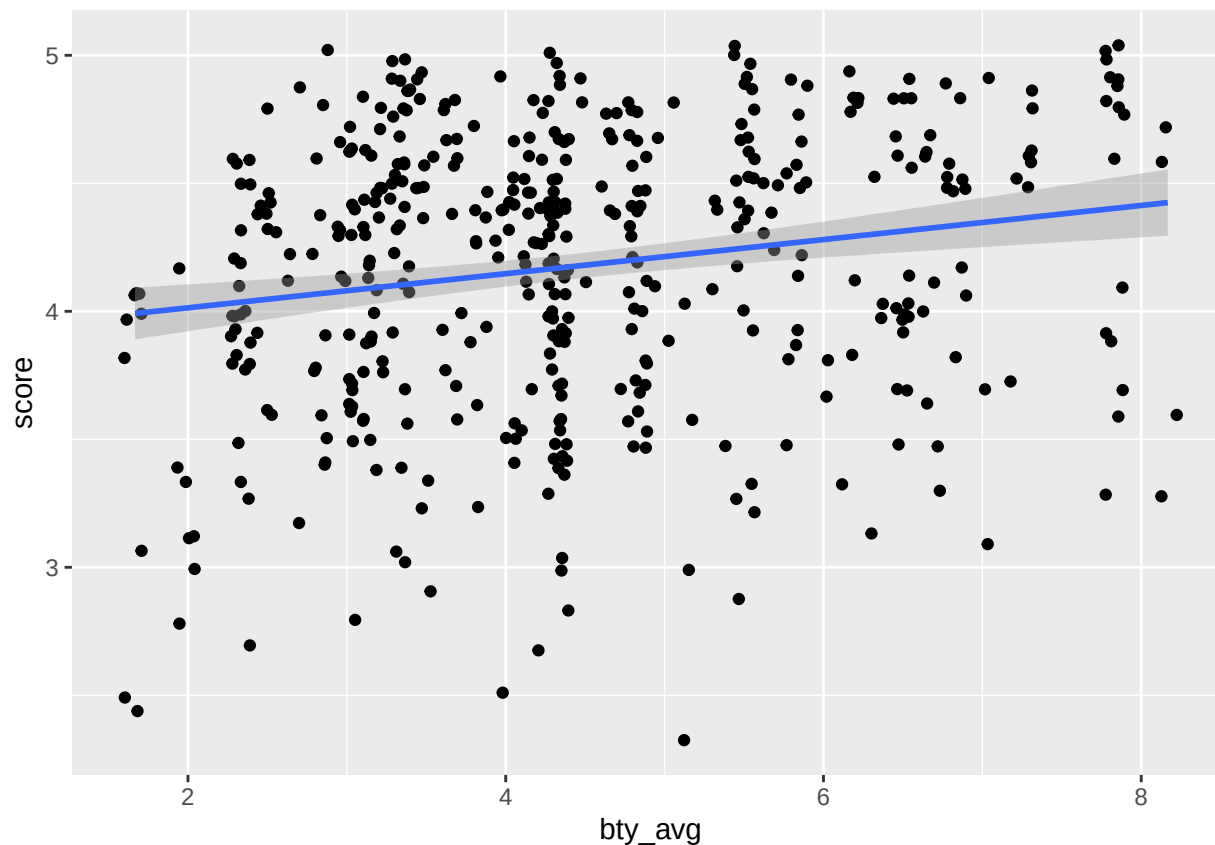
Insert your answer here The initial scatter plot was because they share the same x and y values, causing points to overlap and make it look like a single point. Adding jitter makes it clearer and makes points on top of each other easily distinguishable.

5. Let's see if the apparent trend in the plot is something more than natural variation. Fit a linear model called `m_bty` to predict average professor score by average beauty rating. Write out the equation for the linear model and interpret the slope. Is average beauty score a statistically significant predictor? Does it appear to be a practically significant predictor?

Insert your answer here The equation is: $\hat{y} = 3.8803 + 0.0666 \times \text{bty_avg}$

The slope indicates that each one value increase in average beauty rating will cause the predicted evaluation score increases by about 0.067 points. The pvalue for `bty_avg` is approximately 0.0001, making it statistically significant. Add the line of the best fit model to your plot using the following:

```
ggplot(data = evals, aes(x = bty_avg, y = score)) +  
  geom_jitter() +  
  geom_smooth(method = "lm")
```

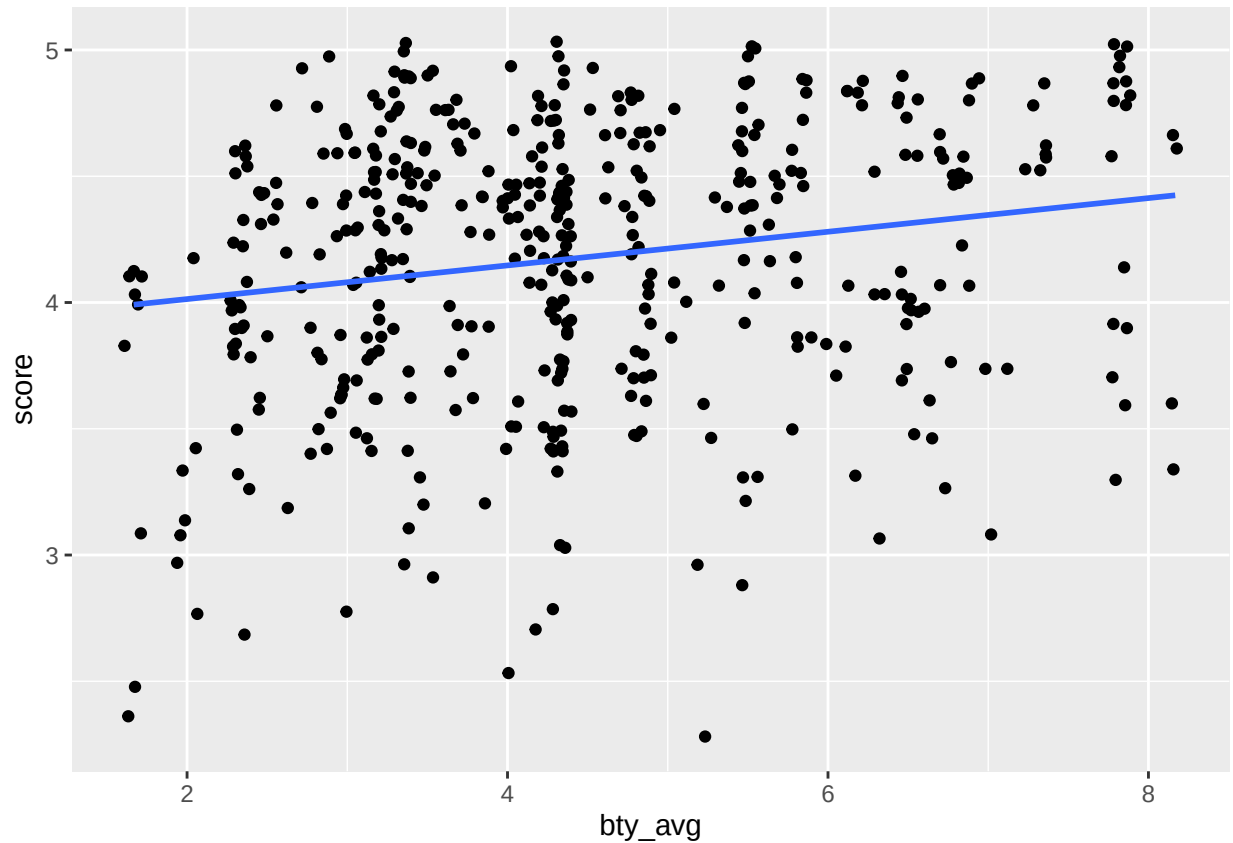


```
m_bty <- lm(score ~ bty_avg, data = evals)
summary(m_bty)
```

```
##
## Call:
## lm(formula = score ~ bty_avg, data = evals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.9246 -0.3690  0.1420  0.3977  0.9309
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.88034    0.07614   50.96 < 2e-16 ***
## bty_avg        0.06664    0.01629    4.09 5.08e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5348 on 461 degrees of freedom
## Multiple R-squared:  0.03502,    Adjusted R-squared:  0.03293
## F-statistic: 16.73 on 1 and 461 DF,  p-value: 5.083e-05
```

The blue line is the model. The shaded gray area around the line tells you about the variability you might expect in your predictions. To turn that off, use `se = FALSE`.

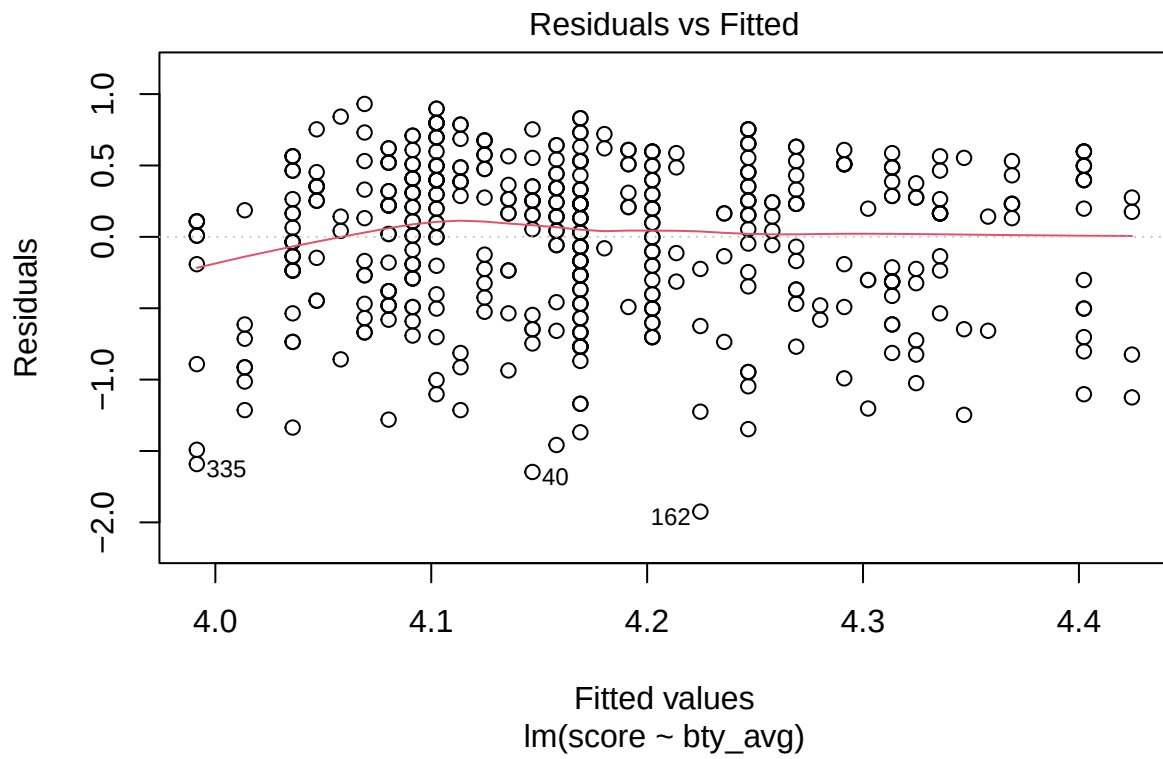
```
ggplot(data = evals, aes(x = bty_avg, y = score)) +  
  geom_jitter() +  
  geom_smooth(method = "lm", se = FALSE)
```



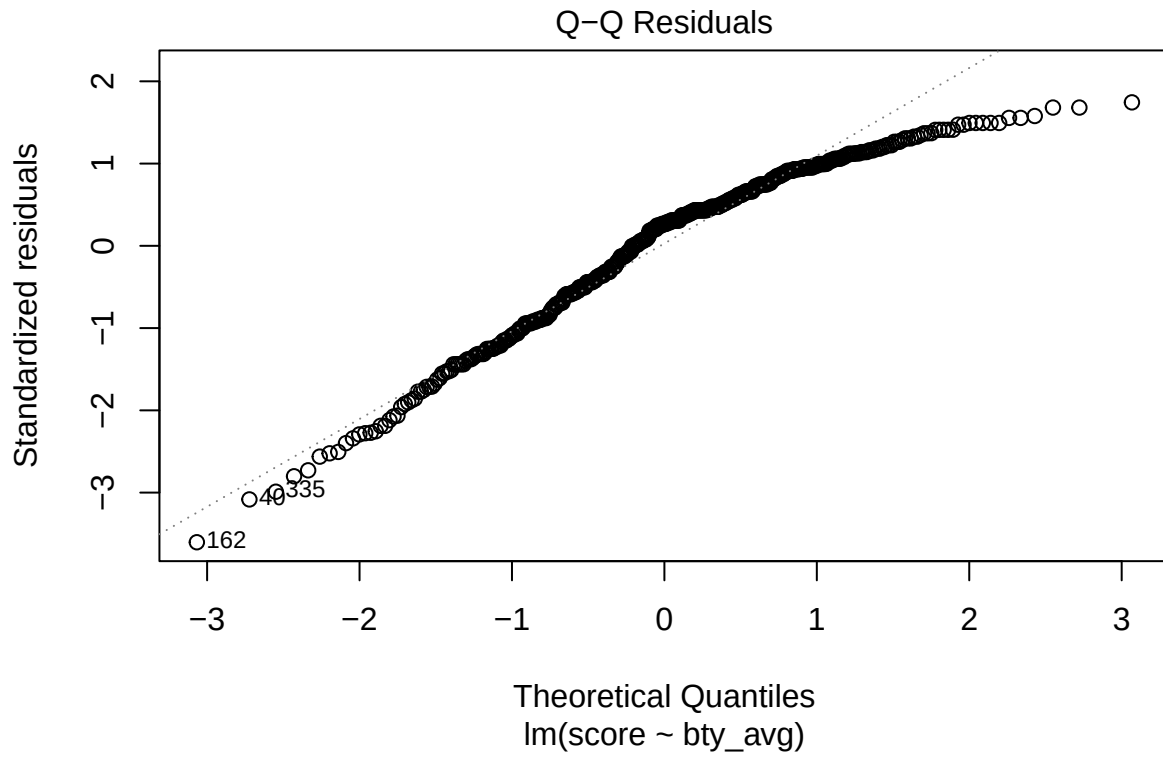
6. Use residual plots to evaluate whether the conditions of least squares regression are reasonable. Provide plots and comments for each one (see the Simple Regression Lab for a reminder of how to make these).

Insert your answer here

```
plot(m_bty, which = 1) # Residuals vs Fitted
```

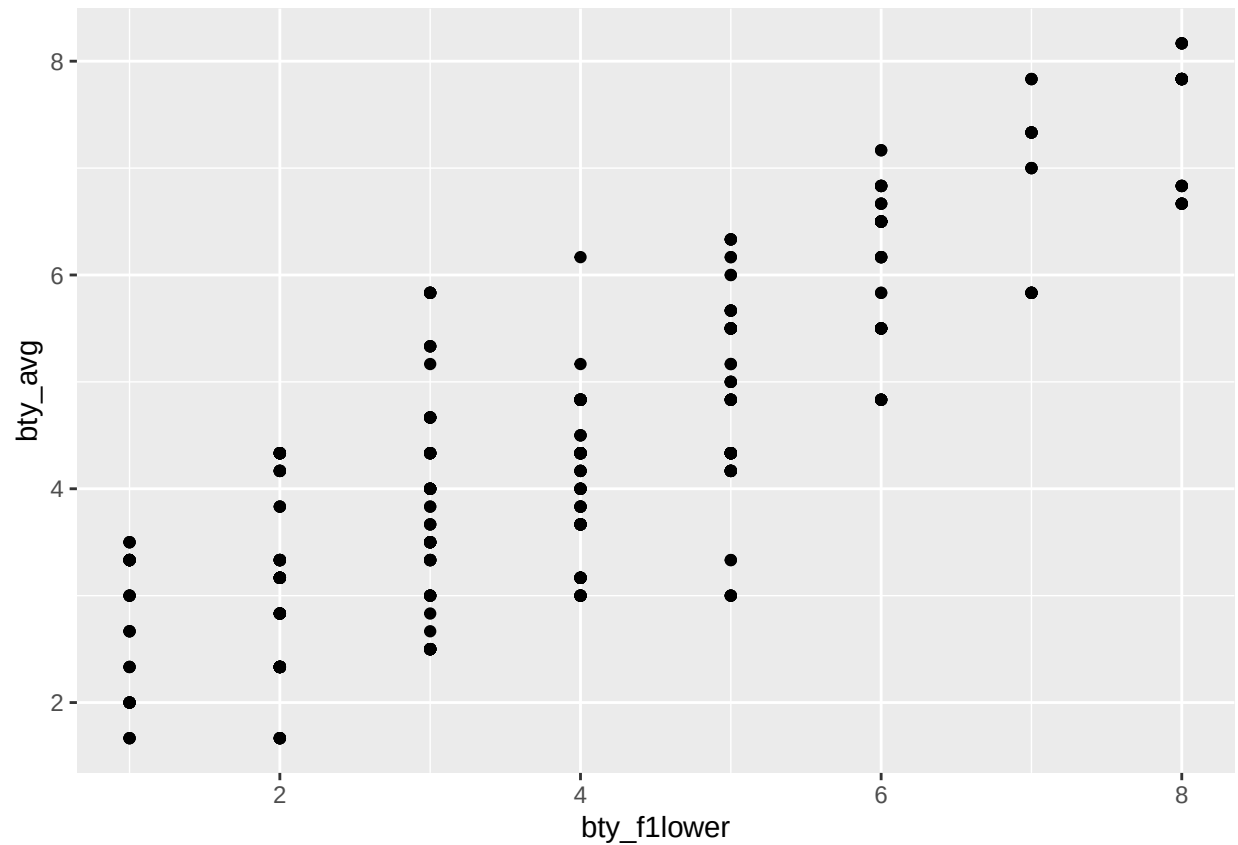
```
plot(m_bty, which = 2) # Normal QQ
```



Multiple linear regression

The data set contains several variables on the beauty score of the professor: individual ratings from each of the six students who were asked to score the physical appearance of the professors and the average of these six scores. Let's take a look at the relationship between one of these scores and the average beauty score.

```
ggplot(data = evals, aes(x = bty_follower, y = bty_avg)) +  
  geom_point()
```

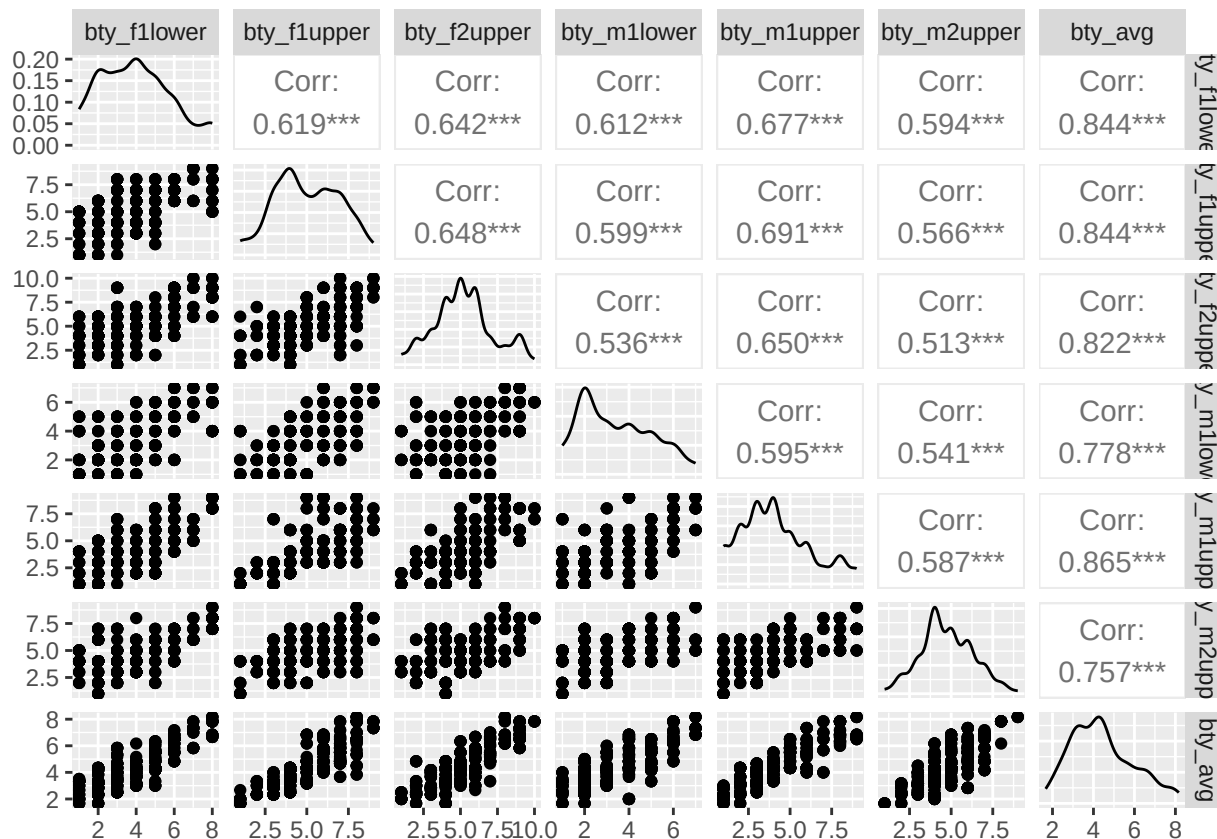


```
evals %>%
  summarise(cor(bty_avg, bty_f1lower))
```

```
## # A tibble: 1 x 1
##   'cor(bty_avg, bty_f1lower)'
##                               <dbl>
## 1                               0.844
```

As expected, the relationship is quite strong—after all, the average score is calculated using the individual scores. You can actually look at the relationships between all beauty variables (columns 13 through 19) using the following command:

```
evals %>%
  select(contains("bty")) %>%
  ggpairs()
```



These variables are collinear (correlated), and adding more than one of these variables to the model would not add much value to the model. In this application and with these highly-correlated predictors, it is reasonable to use the average beauty score as the single representative of these variables.

In order to see if beauty is still a significant predictor of professor score after you've accounted for the professor's gender, you can add the gender term into the model.

```
m_bty_gen <- lm(score ~ bty_avg + gender, data = evals)
summary(m_bty_gen)
```

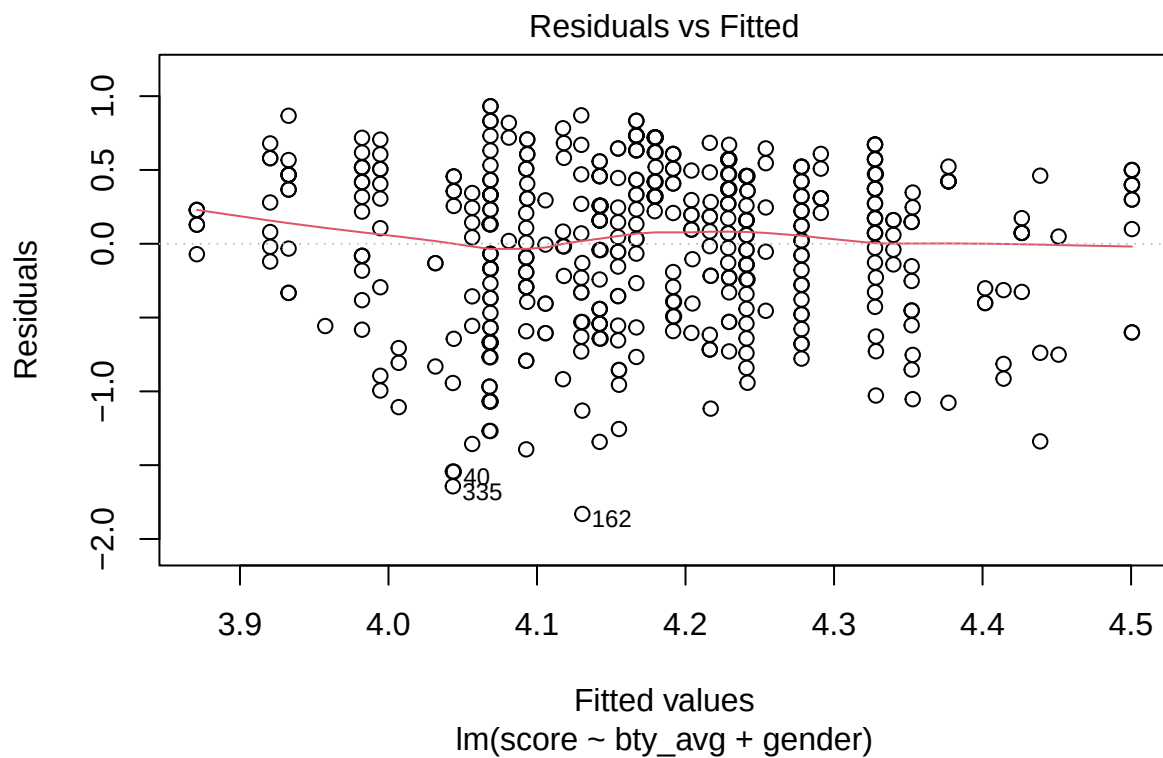
```
##
## Call:
## lm(formula = score ~ bty_avg + gender, data = evals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.8305 -0.3625  0.1055  0.4213  0.9314
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3.74734    0.08466  44.266 < 2e-16 ***
## bty_avg        0.07416    0.01625   4.563 6.48e-06 ***
## gendermale    0.17239    0.05022   3.433 0.000652 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## Residual standard error: 0.5287 on 460 degrees of freedom
## Multiple R-squared:  0.05912,    Adjusted R-squared:  0.05503
## F-statistic: 14.45 on 2 and 460 DF,  p-value: 8.177e-07
```

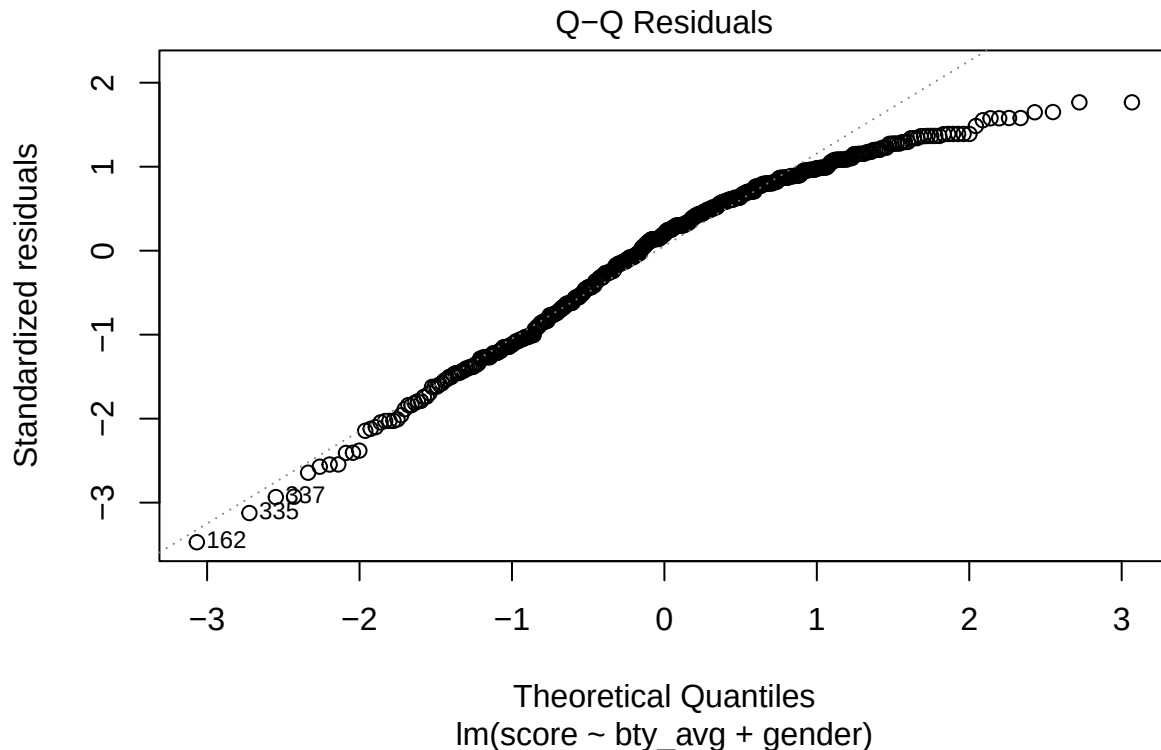
7. P-values and parameter estimates should only be trusted if the conditions for the regression are reasonable. Verify that the conditions for this model are reasonable using diagnostic plots.

Insert your answer here

```
plot(m_bty_gen, which = 1)
```



```
plot(m_bty_gen, which = 2)
```



The plots for `m_bty_gen` look similar to those from `m_bty`. The residual vs fitted doesn't signify a strong pattern and the QQ plot shows a roughly normal distribution with minor left skew.

8. Is `bty_avg` still a significant predictor of `score`? Has the addition of `gender` to the model changed the parameter estimate for `bty_avg`?

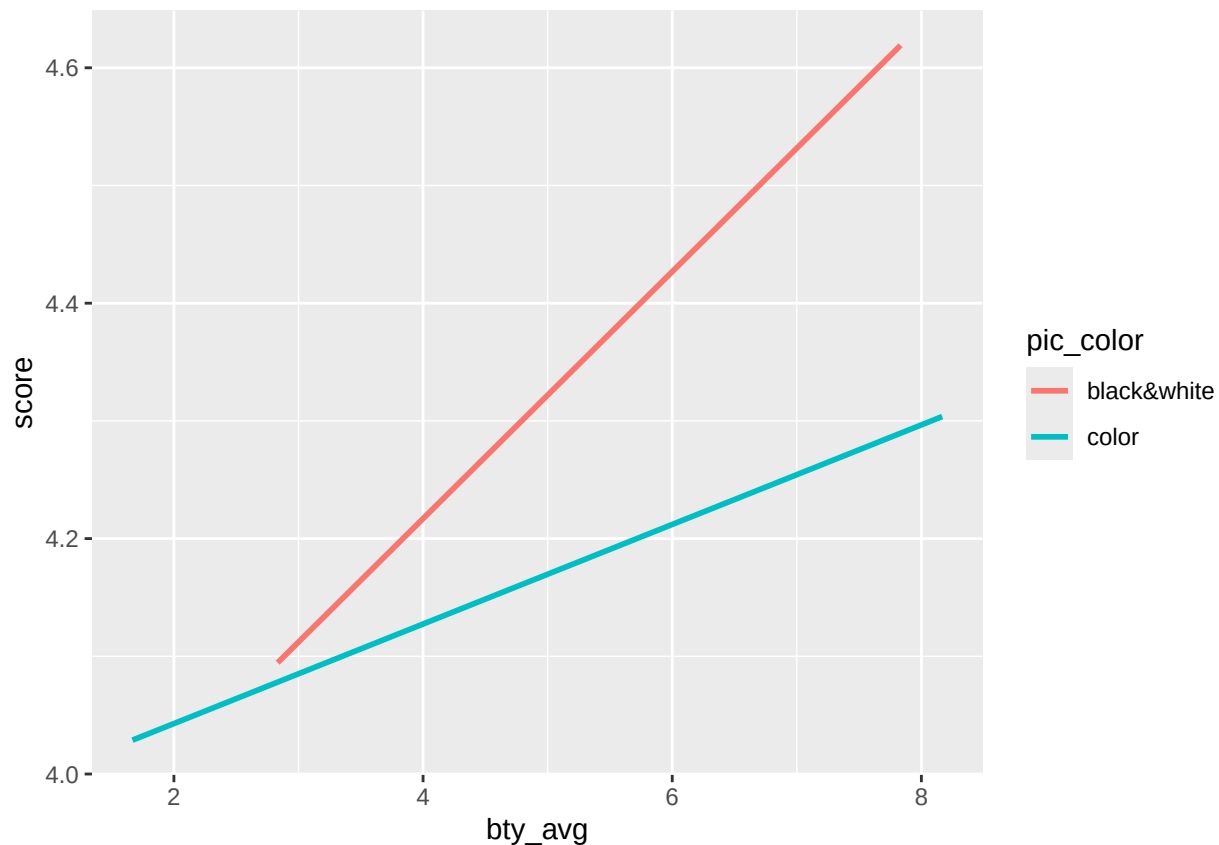
Insert your answer here Yes the `bty_avg` is still a significant predictor of `score`. Adding `gender` slightly increases the estimate which means some sort of confounding effect.

Note that the estimate for `gender` is now called `gendermale`. You'll see this name change whenever you introduce a categorical variable. The reason is that R recodes `gender` from having the values of `male` and `female` to being an indicator variable called `gendermale` that takes a value of 0 for female professors and a value of 1 for male professors. (Such variables are often referred to as "dummy" variables.)

As a result, for female professors, the parameter estimate is multiplied by zero, leaving the intercept and slope form familiar from simple regression.

$$\begin{aligned}\widehat{score} &= \hat{\beta}_0 + \hat{\beta}_1 \times bty_avg + \hat{\beta}_2 \times (0) \\ &= \hat{\beta}_0 + \hat{\beta}_1 \times bty_avg\end{aligned}$$

```
ggplot(data = evals, aes(x = bty_avg, y = score, color = pic_color)) +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE)
```



9. What is the equation of the line corresponding to those with color pictures? (*Hint:* For those with color pictures, the parameter estimate is multiplied by 1.) For two professors who received the same beauty rating, which color picture tends to have the higher course evaluation score?

Insert your answer here

```
yhat= (betahat_0 + betahat_color)+ betahat_bty_avg x btw_avg
```

The decision to call the indicator variable `gendermale` instead of `genderfemale` has no deeper meaning. R simply codes the category that comes first alphabetically as a 0. (You can change the reference level of a categorical variable, which is the level that is coded as a 0, using `therelevel()` function. Use `?relevel` to learn more.)

10. Create a new model called `m_bty_rank` with `gender` removed and `rank` added in. How does R appear to handle categorical variables that have more than two levels? Note that the rank variable has three levels: `teaching`, `tenure track`, `tenured`.

Insert your answer here

```
m_bty_rank <- lm(score ~ bty_avg + rank, data = evals)
summary(m_bty_rank)
```

```
##
## Call:
## lm(formula = score ~ bty_avg + rank, data = evals)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.8713 -0.3642  0.1489  0.4103  0.9525
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.98155    0.09078  43.860 < 2e-16 ***
##  bty_avg        0.06783    0.01655   4.098 4.92e-05 ***
## ranktenure track -0.16070    0.07395  -2.173  0.0303 *
## ranktenured     -0.12623    0.06266  -2.014  0.0445 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5328 on 459 degrees of freedom
## Multiple R-squared:  0.04652,    Adjusted R-squared:  0.04029
## F-statistic: 7.465 on 3 and 459 DF,  p-value: 6.88e-05
```

R handles categorical variables with k levels by creating a k-1 dummy variable.

The interpretation of the coefficients in multiple regression is slightly different from that of simple regression. The estimate for `bty_avg` reflects how much higher a group of professors is expected to score if they have a beauty rating that is one point higher *while holding all other variables constant*. In this case, that translates into considering only professors of the same rank with `bty_avg` scores that are one point apart.

The search for the best model

We will start with a full model that predicts professor score based on rank, gender, ethnicity, language of the university where they got their degree, age, proportion of students that filled out evaluations, class size, course level, number of professors, number of credits, average beauty rating, outfit, and picture color.

11. Which variable would you expect to have the highest p-value in this model? Why? *Hint:* Think about which variable would you expect to not have any association with the professor score.

Insert your answer here `cls_prof` is the variable to have the highest expected p value. The amount of professors teaching the course shouldn't really affect the students rating on the course.

Let's run the model...

```
m_full <- lm(score ~ rank + gender + ethnicity + language + age + cls_perc_eval
             + cls_students + cls_level + cls_profs + cls_credits + bty_avg
             + pic_outfit + pic_color, data = evals)
summary(m_full)
```

```
##
## Call:
## lm(formula = score ~ rank + gender + ethnicity + language + age +
##      cls_perc_eval + cls_students + cls_level + cls_profs + cls_credits +
##      bty_avg + pic_outfit + pic_color, data = evals)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.77397 -0.32432  0.09067  0.35183  0.95036
```



```
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    4.0952141  0.2905277  14.096 < 2e-16 ***
## ranktenure track -0.1475932  0.0820671  -1.798  0.07278 .
## ranktenured     -0.0973378  0.0663296  -1.467  0.14295
## gendermale      0.2109481  0.0518230   4.071 5.54e-05 ***
## ethnicitynot minority 0.1234929  0.0786273   1.571  0.11698
## languagenon-english -0.2298112  0.1113754  -2.063  0.03965 *
## age            -0.0090072  0.0031359  -2.872  0.00427 **
## cls_perc_eval   0.0053272  0.0015393   3.461  0.00059 ***
## cls_students    0.0004546  0.0003774   1.205  0.22896
## cls_levelupper   0.0605140  0.0575617   1.051  0.29369
## cls_profssingle -0.0146619  0.0519885  -0.282  0.77806
## cls_creditsone credit 0.5020432  0.1159388   4.330 1.84e-05 ***
## bty_avg         0.0400333  0.0175064   2.287  0.02267 *
## pic_outfitnot formal -0.1126817  0.0738800  -1.525  0.12792
## pic_colorcolor  -0.2172630  0.0715021  -3.039  0.00252 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.498 on 448 degrees of freedom
## Multiple R-squared:  0.1871, Adjusted R-squared:  0.1617
## F-statistic: 7.366 on 14 and 448 DF,  p-value: 6.552e-14
```

12. Check your suspicions from the previous exercise. Include the model output in your response.

Insert your answer here cls_profs has the highest p value of 0.77 confirming our suspicions from the previous exercise.

13. Interpret the coefficient associated with the ethnicity variable.

Insert your answer here The coefficient for ethnicity variable is around 0.123 which means that holding all other variables constant, professors who are not members of a minority group are expected to receive evaluation scores about 0.12 points higher than minority professors on average.

14. Drop the variable with the highest p-value and re-fit the model. Did the coefficients and significance of the other explanatory variables change? (One of the things that makes multiple regression interesting is that coefficient estimates depend on the other variables that are included in the model.) If not, what does this say about whether or not the dropped variable was collinear with the other explanatory variables?

Insert your answer here

```
m_full2 <- lm(score ~ rank + gender + ethnicity + language + age + cls_perc_eval
              + cls_students + cls_level + cls_credits + bty_avg
              + pic_outfit + pic_color, data = evals)
summary(m_full2)
```

```
##
## Call:
## lm(formula = score ~ rank + gender + ethnicity + language + age +
```

```
##      cls_perc_eval + cls_students + cls_level + cls_credits +
##      bty_avg + pic_outfit + pic_color, data = evals)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -1.7836 -0.3257  0.0859   0.3513   0.9551
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      4.0872523   0.2888562   14.150 < 2e-16 ***
## ranktenure track  -0.1476746   0.0819824   -1.801 0.072327 .
## ranktenured       -0.0973829   0.0662614   -1.470 0.142349
## gendermale        0.2101231   0.0516873    4.065 5.66e-05 ***
## ethnicitynot minority 0.1274458   0.0772887    1.649 0.099856 .
## languagenon-english -0.2282894   0.1111305   -2.054 0.040530 *
## age              -0.0089992   0.0031326   -2.873 0.004262 **
## cls_perc_eval      0.0052888   0.0015317    3.453 0.000607 ***
## cls_students       0.0004687   0.0003737    1.254 0.210384
## cls_levelupper     0.0606374   0.0575010    1.055 0.292200
## cls_creditsone credit 0.5061196   0.1149163    4.404 1.33e-05 ***
## bty_avg            0.0398629   0.0174780    2.281 0.023032 *
## pic_outfitnot formal -0.1083227   0.0721711   -1.501 0.134080
## pic_colorcolor     -0.2190527   0.0711469   -3.079 0.002205 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4974 on 449 degrees of freedom
## Multiple R-squared:  0.187, Adjusted R-squared:  0.1634
## F-statistic: 7.943 on 13 and 449 DF, p-value: 2.336e-14
```

Dropping cls_prof causes minimal changes in the other coefficients and the other variables significant values. This indicates a low colinearity.

15. Using backward-selection and p-value as the selection criterion, determine the best model. You do not need to show all steps in your answer, just the output for the final model. Also, write out the linear model for predicting score based on the final model you settle on.

Insert your answer here

```
m_final <- lm(score ~ gender + ethnicity + language + age + cls_perc_eval
              + cls_credits + bty_avg + pic_color, data = evals)
summary(m_final)

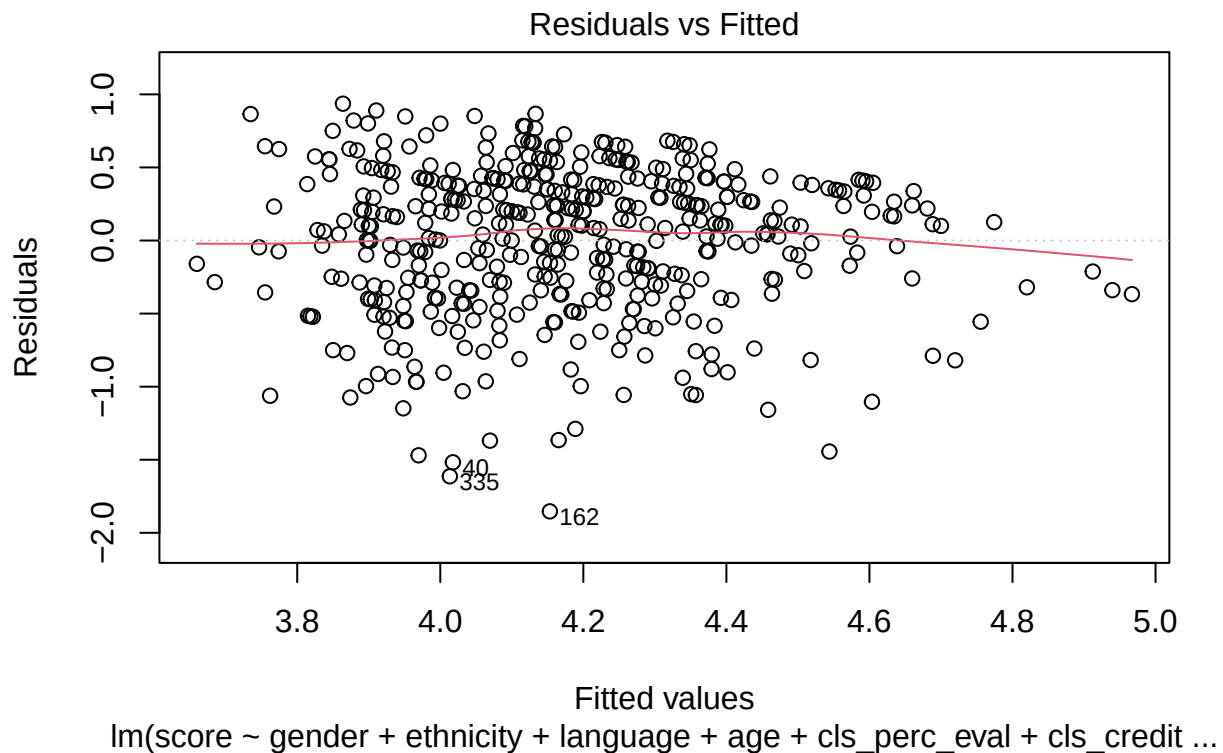
##
## Call:
## lm(formula = score ~ gender + ethnicity + language + age + cls_perc_eval +
##      cls_credits + bty_avg + pic_color, data = evals)
##
## Residuals:
##      Min        1Q    Median        3Q        Max
## -1.85320 -0.32394  0.09984   0.37930   0.93610
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.771922   0.232053  16.255 < 2e-16 ***
## gendermale        0.207112   0.050135   4.131 4.30e-05 ***
## ethnicitynot minority 0.167872   0.075275   2.230 0.02623 *
## languagenon-english -0.206178   0.103639  -1.989 0.04726 *
## age              -0.006046   0.002612  -2.315 0.02108 *
## cls_perc_eval      0.004656   0.001435   3.244 0.00127 **
## cls_creditsone credit 0.505306   0.104119   4.853 1.67e-06 ***
## bty_avg           0.051069   0.016934   3.016 0.00271 **
## pic_colorcolor    -0.190579   0.067351  -2.830 0.00487 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4992 on 454 degrees of freedom
## Multiple R-squared:  0.1722, Adjusted R-squared:  0.1576
## F-statistic: 11.8 on 8 and 454 DF, p-value: 2.58e-15
```

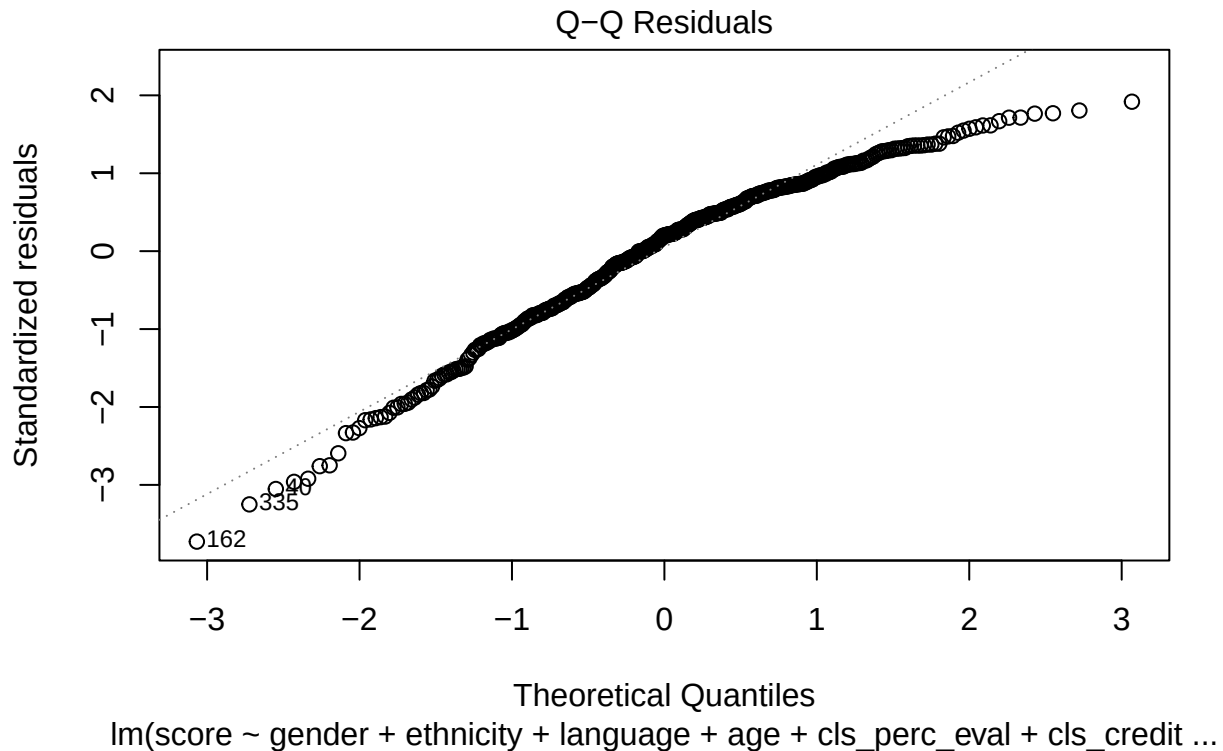
16. Verify that the conditions for this model are reasonable using diagnostic plots.

Insert your answer here

```
plot(m_final, which = 1)
```



```
plot(m_final, which = 2)
```



17. The original paper describes how these data were gathered by taking a sample of professors from the University of Texas at Austin and including all courses that they have taught. Considering that each row represents a course, could this new information have an impact on any of the conditions of linear regression?

Insert your answer here This information can have an impact for the independent conditions. Each row is a course and some professors can appear multiple times. Some courses can be non independent observations.

18. Based on your final model, describe the characteristics of a professor and course at University of Texas at Austin that would be associated with a high evaluation score.

Insert your answer here Based on my final model a professor associated with a higher evaluation score would tend to be male, not a minority, and would have their degree from an english speaking school, and younger.

19. Would you be comfortable generalizing your conclusions to apply to professors generally (at any university)? Why or why not?

Insert your answer here

No i would not be comfortable generalizing broadly. The data comes exclusively from one institution with a specific student population and culture in that school. These factors that affects evaluation culture varies in every instituion and can all have their different systems.